CF-GCN: Graph Convolutional Network for Change Detection in Remote Sensing Images

Wei Wang, Cong Liu, Guanqun Liu, and Xin Wang

***Abstract*— The remote sensing image change detection methods based on deep learning have made great progress. However, many convolutional neural network (CNN)-based methods persistently face challenges in connecting long-range semantic concepts because of their limited receptive fields. Recently, some methods that combine transformers effectively extract global information by modeling the context in the temporal and spatial domains have been proposed to solve the problem, but they still suffer from both the incorrect identification of "non-semantic changes" and the incomplete and irregular boundary extraction due to the deterioration of local feature details. In response to these inquiries, we propose a novel network, coordinate space and feature interaction-graph convolutional network (CF-GCN), based on graph convolutional structures for change detection. Specifically, in the encoder and decoder of the network, different projection strategies are employed to construct a coordinate space graph convolution network (GCN_C) and feature interaction graph convolution network (GCN_F). The boundary perception module (BPM) extracts spatial boundary features of shallow layers and enhances boundary perception ability during graph-based information propagation, effectively suppressing the tendency of image boundary information to gradually smooth out. At the same time, the knowledge review module (KRM) is utilized to form knowledge complementarity between key layers of the network, effectively mitigating the propagation of erroneous knowledge in the deep network. On the LEVIR-CD dataset, the intersection over union (IoU) score of CF-GCN is 83.41%, which is 0.35% and 0.39% higher than ChangeStar and DMINet, respectively. On the WHU-CD dataset, the *F*1 and IoU are as high as 91.83% and 84.90%, which are significantly better than other state-of-the-art (SOTA) networks. The experimental results show that, in addition to CNN and Transformer, the graph-convolution structure approach is expected to be another major research direction for performing fully supervised change detection. Our code and pretrained models will be available at https://github.com/liucongcharles/CF-GCN.**

***Index Terms*— Boundary perception, change detection, graph convolutional network (GCN), remote sensing images.**

I. Introduction

THE purpose of remote sensing image change detection is to detect the difference pixels between bi-temporal remote sensing images acquired from the same area at different times. It is widely used in disaster assessment [1], [2], [3], urbanization detection [4], [5], [6], and environmental monitoring [7], [8]. With the development of sensors, the temporal and spatial resolution has gradually improved, enhancing the ability to acquire multitarget remote sensing images. Since high-resolution remote sensing (HRSS) images can provide rich spatial information and fine surface details, detecting changes in them is important for fine-grained applications such as precision agriculture and urban renewal [9].

HRSS images have obvious geometric features, complex texture information, and limited spectral information. Objective factors such as different acquisition times and differences in sensor noise angles can easily cause spectral variations and loss of spatial information [9]. In our study, "non-semantic change" are changes caused by camera motion, sensor noise, light change, or human subjective factors [10]. As shown in Fig. 1, the red boxes are some "non-semantic change" regions that affect the detection performance. In the first row of Fig. 1, building shadows are recognized as change regions. In the second row, the presence of cars and roads as "non-semantic changes" can significantly affect the change detection results. In the third row, there is a significant difference in spectral features due to different roof colors in seasonal and lighting conditions, which can be easily detected as change regions. The presence of large areas of artificial concrete floors and buildings in the fourth row is also subjectively specified as an invariant region, leading to uncertainty in the discrimination process. How to effectively extract and acquire rich feature information, eliminate the interference of "non-semantic changes" to improve the detection accuracy is an important issue in the current HRSS image change detection task.

Traditional change detection methods, such as image differencing methods [11], change vector analysis [12], and principal component analysis [13], compare pixel values in order to generate difference maps, and then produce change maps based on thresholds. These methods focus only on the variations between individual pixels, ignore spatial contextual information, and face many problems such as determining empirical thresholds. Such methods make it difficult to accurately predict the change region of pixels in complex scenes.

In recent years, deep convolutional neural networks (CNN) have outperformed traditional methods in multiple fields, including image recognition, image segmentation, and natural language processing [10], [14], [15], [16], [17]. Meanwhile, deep CNN are also widely used in remote sensing image

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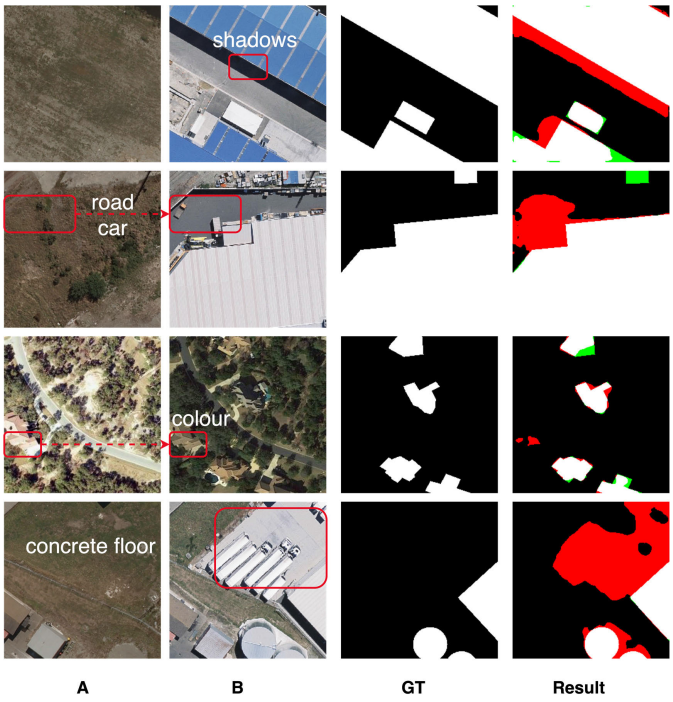


Fig. 1. Variety of "non-semantic changes" affects change detection results. A, B are images of bi-temporal images. GT is the ground truth. The results, in order from top to bottom, are Siam-Di, Bit, ChangeFormer, and ChangeStar. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

change detection. Mainstream deep learning methods can be divided into three categories as follows.

1) Pure convolutional methods [18], [19], [20], [21], [22]. They expand the receptive field by stacking more convolutional layers, and apply attention mechanisms to extract features from multiple dimensions such as space or channels [18], [19]. Additionally, they use dense connectivity to extract multiscale information [10], [23], and apply clustering algorithms to generate change maps by clustering bi-temporal data into change regions and invariant regions [24], [25]. However, pure convolutional methods cannot capture long-distance relationships between pixels in an image due to their limited receptive fields [14].
2) The methods combined with the transformer feature extraction [14], [26], [27], [9], [28]. They effectively model the long-distance context information in bi-temporal images and improve the feature representation for the original pixel space. These methods usually rely on the global representation of the image. And their ability for performing relational reasoning for "non-semantic changes" in complex remote sensing scenes needs to be improved, because of higher computational costs.
3) Using graph convolution methods [29], [30]. These methods capture the relationships between pixels in HRRS images and obtain more comprehensive feature information. The feature representation capability of graphs is utilized to explore the semantic similarity of unlabeled pixels in an image to generate reliable pseudo-labels in unsupervised or semi-supervised tasks. Due to the complexity of graph computation, many tasks use graph convolution only as an auxiliary feature extractor to enhance the coupling of information between bi-temporal images [31]. Especially in fully supervised change detection tasks, the trade-off between model complexity and detection accuracy is a problem that needs to be addressed for graph convolution applications.

Considering the specific characteristics of HRRS images, including complex scenes and a large number of "non-semantic changes," we propose a new coordinate space and feature interaction-graph convolutional network (CF-GCN). In the graphical representation of images, each node represents a local region of the image, while the edges between nodes capture the spatial relationships within the local neighborhood. By using this graph representation, the spatial context of each region and the relationship with neighboring regions are captured, enabling effective inference of changed regions. Because most networks focus more on the semantic information in deep layers, it is easy to ignore the detailed information (boundary features) in shallow layers. While the ability of boundary knowledge to acquire spatial features is crucial in pixel-level prediction tasks [32], [33], [34]. In order to prevent boundary smoothing while deepening the network and propagating the incorrect classification results in deeper networks, we propose the boundary perception module (BPM) and knowledge review module (KRM) modules to improve the network discrimination by means of differential generation and multilevel aggregation.

The main contributions of this article are as follows.

1) In order to eliminate the interference of "non-semantic changes," a new GCN Encoder and GCN Decoder are proposed. These two modules are constructed in a way different from most graph structures based on non-local blocks to build networks of graph convolution, which can effectively improve the inference ability of graph structure for semantic relationships.
2) For the purpose of retaining the complete boundary features, the BPM and KRM are proposed. The tendency of boundary features to be gradually smoothed as the network deepens is effectively suppressed through BPM. KRM mitigates the propagation of classification errors at different stages of the network caused by the diversity of target object features, forming knowledge complementarity between key layers.
3) To verify the effectiveness of the proposed CF-GCN based on the graph convolution framework from multiple perspectives in remote sensing image change detection tasks, extensive experiments were carried out on two classic datasets. Through quantitative and qualitative comparisons with other state-of-the-art (SOTA) models, CF-GCN not only reflects obvious advantages in multiple detection metrics, but also achieves a better trade-off between model complexity and accuracy.

The rest of this article is organized as follows. Section II briefly reviews related work on deep-learning-based CNN, transformer, and GCNs. In Section III, the proposed change

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detection method is described in detail. Experiments and discussions are presented in Section IV. Finally, Section V summarizes the article.

II. Related Work

A. Change Detection Model Based on CNN or Transformer

Many change detection methods are based on CNNs, due to their powerful feature representation capabilities. These models enhance the semantic representation of the network by changing the network structure, optimizing the loss function, adding attention mechanisms, and so on. The change detection method based on Siamese networks was proposed for the first time [35]. Most of the current change detection methods depend on the Siamese network structure, which allows the bi-temporal images to be processed simultaneously. Three fully convolutional network (FCN)-based fully CNN structures have been proposed for end-to-end training [36]. FC-EF concatenated bi-temporal images as inputs. Both FC-Siam-Conc and FC-Siam-Di used Siamese network structures to process bi-temporal images directly. DTCDSCN [37] introduced the attention mechanism into the change detection task to learn more image features through the dual-dimensional attention mechanism in channel and space. STANet [18] proposed the basic spatial-temporal attention module (BAM) and the pyramid spatial-temporal attention module (PAM) to capture long-range, rich spatial-temporal relationships. By using dense connections, SNUNet [10] took advantage of the network's shallow low-order features with deep high-order features, fusing features from several network layers. BIT [14] first introduced a transformer to the change detection task. With the powerful receptive field of the transformer, the context in the spatial-temporal domain was efficiently modeled. ChangeFormer [26] unified a hierarchically structured transformer encoder with a multilayer perception (MLP) decoder in a Siamese network architecture to efficiently render multiscale long-range details required for accurate change detection tasks. ICIFNet [38] used an intrascale cross-interaction and interscale feature fusion to acquire local and global features to solve the misalignment problem caused by gradual down-sampling operations in the backbone network. ChangStar [39], utilized changeMix branches to temporarily exchange bi-temporal images and provided the generation of inductive biases in the network architecture to enhance the feature extraction capability of the backbone network. DMINet [40] proposed a cross-period JointAttention (JointAtt) block to guide the global feature distribution of each input, stimulating the coupling of information between the in-layer representation and the mean.

The above methods have improved the capability of change detection models to some extent. However, due to the inherent localization of convolutional network perceptual fields and the high computational cost of transformers, those methods cannot effectively balance accuracy and model complexity and have insufficient inference capability for "non-semantic change" regions. Therefore, the performance of their models is limited. Different from the previous approaches, this article mainly explores the potential of graph convolutional architectures for change detection tasks.

B. Development and Applications of GCN

Methods based on GCNs [41], [42], [43], [44], [45] have become increasingly popular in recent years. GCN provides a large number of solutions for non-Euclidean data processing and has proven to be a very effective method for relational inference. Because semantic information in images is a type of non-Euclidean data, so GCN is suitable for computer vision tasks. Non-local networks [42] learned the affinity map between all pixels in an image or video frame by capturing the dependencies of remote pixels. This enabled the network to effectively expand the receptive field and be applied in tasks such as video classification and target detection. GraphFCN [43] used an intermediate feature layer to construct a graph structure to pass into GCN, then classified each node in the graph to change the classification problem for each pixel to a classification problem for the graph nodes. DGCNet [44] used different projection strategies to model the spatial relationships between pixels and the interdependencies of channel dimensions. So that the long-distance interaction of contextual information can be achieved. BI-GCN [45] made use of both region and boundary information when performing graph-based information dissemination. The semantic reasoning of boundary region pixels is enhanced by increasing the representational capabilities for graphs to model and infer different categories of boundary-aware region correlations.

In semi-supervised or unsupervised change detection tasks, DCVA [29] used graph convolution to efficiently process dispersed labeled samples from a pool of unlabeled samples for capturing spatial and spectral information from bi-temporal images to improve change detection performance. GMCD [30] focused on capturing the short- and long-range contextual patterns at the feature map level to extract information completely from changed and unchanged areas. FD-MCD [46] computed local frequency consistency measures in the Fourier domain to determine the local structure difference. It constructed non-local structure relationship graphs for prechange and post-change images, respectively, and obtained difference maps by fusing the local and non-local structures. In a fully supervised change detection task, BGINet [31] proposed a unified self-focusing mechanism for bi-temporal images. This approach enhanced the coupling of information between two temporal images while suppressing task-irrelevant interference.

Utilizing the idea of graph convolution, which had achieved great success in many tasks of computer vision, we proposed a novel network, named CF-GCN, for performing remote sensing image change detection. Unlike most of the graph convolution methods based on non-local blocks mentioned above, we have constructed a spatial interaction graph convolution based on the concept of non-local blocks and added a feature interaction branch for channel interaction. The experimental results show that our network achieved higher accuracy than other SOTA networks in terms of inferring "non-semantic change" regions.

III. Methodology

The overall structure of the network is shown in Fig. 2(a). CF-GCN starts with the backbone network, where the

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bitemporal images are input into the Siamese backbone network to obtain the original features of the input images. The feature maps are fused and input to CF-GCN for encoding operation to obtain semantic information in the deep network. After decoding, the model passes through the prediction head to generate pixel-level change predictions. CF-GCN has four main modules as follows.

1) Graph encoder, which is used to establish long-distance dependencies between pixel nodes.
2) BPM, which is used to obtain boundary information at multiple scales of the shallow network.
3) Graph decoder further identifies "semantic changes" in the image by comparing the original features and fusion features, combining them with boundary information, and excluding the interference of "non-semantic changes."
4) KRM module is used to complement knowledge between the three key layers of the network to further enhance network performance.

A. Graph Encoder

The weighted sum of the central and neighboring nodes is calculated by standard convolution. It is used for integrating information for the neighbor nodes. In standard convolution, information is only exchanged among positions in a small neighborhood defined by the filter size. To capture long-distance dependencies, multiple convolutional layers must be stacked [40]. Graph convolution also contains the definition of the neighborhood used in standard convolution. By defining edges between nodes in the graph, long-distance information exchange in a single network layer can be accomplished. In Graph convolution [41], [42], [43], [44], [45], the formal definition is given as

$$X^{\wedge} = \sigma(AXW) \tag{1}$$

where $\sigma(\cdot)$ is the non-linear activation function, A is the adjacency matrix that characterizes the neighborhood relations between the graph's nodes, and W is the weight matrix.

The definition and structure of the graph are critical in determining information propagation. The Graph Encoder constructs an orthogonal graph space using two different graph projection strategies to learn the different features in the change detection task, as shown in Fig. 2(c).

The internal structure of the Graph Encoder is implemented as follows.

1) Coordinate Space Graph Convolution: The coordinate space graph convolution network (GCN_C) projects the input feature maps $X \in \mathbb{R}^{H\times W\times D}$ of the backbone network into a new coordinate space $\mathbf{\Omega}$. This is accomplished through a downsampling operation at a rate of d, converting the input feature X to a new feature $X_s \in \mathbb{R}^{((H\times W)/d^2)\times D}$. In the coordinate space $\mathbf{\Omega}$, three learnable linear transformations $\mu(\cdot)$, $\varphi(\cdot)$, and $\phi(\cdot)$ are used to generate the components of the graph convolution, formulated as

$$M_s = \zeta\left(\varphi(X_s), \phi(X_s)^T\right)\mu(X_s)W_s \tag{2}$$

where $\mu(\cdot)$, $\varphi(\cdot)$, $\phi(\cdot)$ are implemented by different 1×1 convolutions, $\zeta(\cdot)$ is the dot-product operation. $\zeta(\varphi(X_s), \phi(X_s)^T)$ constitutes the adjacency matrix A in the definition of the graph convolution, and W_s is the random parameterization matrix.

After completing the graph convolution inference, the new features are projected back into the original coordinate space $\mathbb{R}^{H\times W\times D}$ to make them compatible with a regular CNN. This process can be written as

$$X_s = \psi(X \times \text{interp}(M_s)) \tag{3}$$

where interp$(\cdot)$ is the nearest neighbor interpolation algorithm used as the reprojection operation. M_s is projected to the original input image size $H \times W$ with input feature X as the reference. interp$(\cdot)$ is matrix multiplied with X to enhance the original features. Then, the channel dimension of X_s is converted to the D dimension by $\psi(\cdot)$, which is a 1×1 convolution.

2) Feature Interaction Graph Convolution: The feature interaction graph convolution network (GCN_F) projects the input features $X \in \mathbb{R}^{H\times W\times D}$ to the feature interaction space $\mathcal{H}$ through the projection function is defined as

$$M_{\mathcal{H}} = \alpha(X)\beta(X), \quad M_{\mathcal{H}} \in \mathbb{R}^{D_2\times D_1} \tag{4}$$

where $\alpha(X) \in \mathbb{R}^{N\times D_1}$, $\beta(X) \in \mathbb{R}^{N\times D_2}$, $N = H \times W$. $\alpha(\cdot)$, $\beta(\cdot)$ implemented by a 1×1 convolution. The matrix multiplication is performed after the dimensionality reduction of X, respectively. The new feature $M_{\mathcal{H}}$ aggregates the information from multiple channels.

The feature $M_{\mathcal{H}}$ is used as the node information of a fully connected graph. Through graph convolution, each node interacts with all the underlying features to update its state for learning the relationship between the node and the global features. The process can be expressed as follows:

$$H_f = (I + A_{\mathcal{H}})M_{\mathcal{H}}W_{\mathcal{H}} \tag{5}$$

where the unit matrix I is used to construct the adjacency matrix $I + A_{\mathcal{H}}$ through Laplacian smoothing [44], [47], [48], which is used to propagate features among nodes. The parameter matrix $W_{\mathcal{H}}$ is used for feature interaction in graph convolution. The parameters of $A_{\mathcal{H}}$ and $W_{\mathcal{H}}$ are randomly initialized and optimized using the gradient descent algorithm.

The feature H_f is projected back to the original space, $X_f \in \mathbb{R}^{N\times D}$, which is achieved through matrix multiplication with X. The channel dimension is transformed into D dimensions through operation $\psi(\cdot)$. X_f can be described as follows:

$$X_f = \psi(X \times H_f). \tag{6}$$

The final output feature $\widetilde{X}$ of the Graph Encoder is

$$\widetilde{X} = X + X_s + X_f. \tag{7}$$

The coordinate spatial graph convolution captures the relationships between nodes at different positions along the spatial dimension. On the other hand, the GCN_F obtains contextual information along the channel dimension in the feature map. These two operations complement each other and are combined with the original feature X through a residual structure [49]. This enhances the semantic representation ability of the encoding module.

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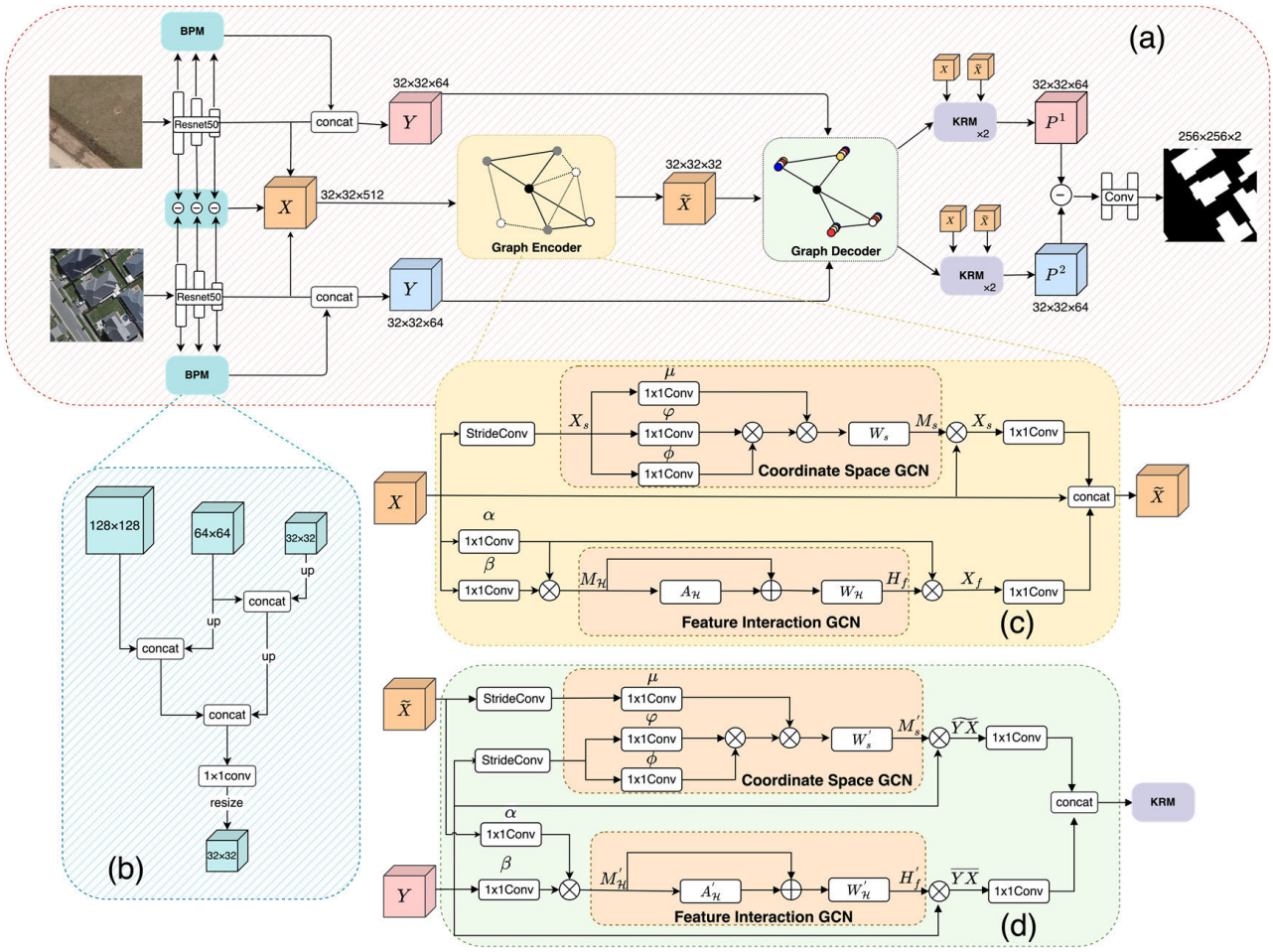


Fig. 2. (a) Overall network architecture of CF-GCN. (b) BPM, designed specifically for addressing the boundary misalignment issue in change detection, effectively aligns the boundary region. (c) Graph Encoder module models the relationships between pixel nodes through two dimensions: the coordinate space graph and the feature interaction graph, enabling the network to capture both spatial and semantic information. (d) Graph Decoder module effectively identifies "non-semantic changes" by comparing the difference between fused features and original features.

B. Boundary Perception Module

The deep and shallow features from different layers of the network complement each other. The deep features contain rich spatial semantic information, while the shallow features retain sufficient spatial boundary information [50], [51]. However, most networks require downsampling for input images due to the limitation of memory and speed. Down-sampling might result in the loss of many boundary information, leading to gradually smoothed boundary regions or other problems, like incomplete prediction areas and irregular boundaries. To address this issue, the BPM is proposed to aggregate shallow image-level features of the backbone network, as shown in Fig. 2(b). During the training process, the boundary information is fed into the Graph Decoder along with the original feature maps, and the differential boundary information is fed into the encoder to help the model predict boundary regions more accurately. By introducing this auxiliary module, the model can better utilize the spatial information of the input image, thereby improving its performance and robustness. The output is defined as

$$Y_f = \tau(Y_3, \mathrm{up}(\tau(Y_2, \mathrm{up}(Y_1)))) \tag{8}$$

where $Y_1 \in \mathbb{R}^{128\times128\times D_{y_1}}$, $Y_2 \in \mathbb{R}^{64\times64\times D_{y_2}}$, $Y_3 \in \mathbb{R}^{32\times32\times D_{y_3}}$ are the output features of the first three layers of the backbone network, $\tau(\cdot)$ represents the channel fusion operation, and $\mathrm{up}(\cdot)$ represents the upsampling twice. Y_f is sampled by the nearest neighbor interpolation algorithm $\mathrm{interp}(\cdot)$ to obtain with the same size as the feature $\widetilde{X}$.

C. Graph Decoder

The Graph Decoder constructs a dual-dimensional graph convolution structure that combines encoding features with original features, as shown in Fig. 2(d). Establishing a dependency relationship between the original features output of the backbone network and the fused features outputted by the Graph Encoder can enhance the ability of extracting semantic information from the images. This approach effectively identifies "non-semantic changes" in change detection tasks. In the feedforward network, a single-temporal feature map Y is obtained by fusing boundary features Y_e with the output features X of the backbone network. The channel fusion operation is denoted by $\mathrm{cat}(\cdot)$ and the formula is shown as $Y = \mathrm{cat}(Y_e, X)$. In the GCN_C of Graph Decoder, the input feature Y is projected as the new feature Y_s. A graph convolution is constructed with Y_s as the base. The detailed operation is given in

$$M_s' = \zeta(\varphi(Y_s), \phi(Y_s)^T)\mu(\widetilde{X}_s)W_s' \tag{9}$$

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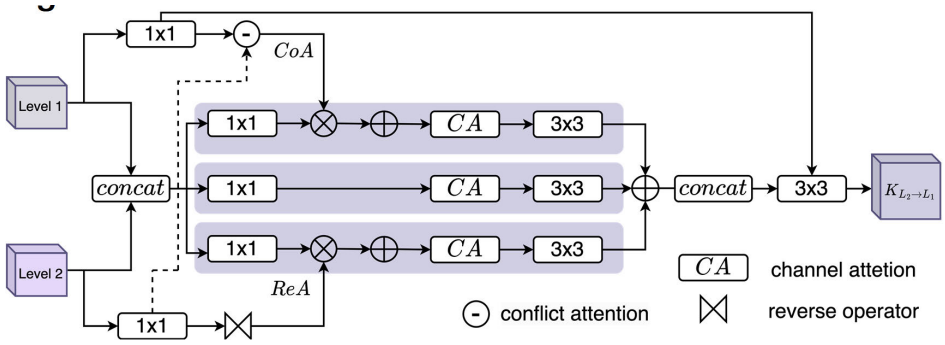


Fig. 3. Detailed structure of the KRM.

where $\zeta(\varphi(Y_s), \phi(Y_s)^T)$ represents the adjacency matrix, $\mu(\widetilde{X}_s)$ is the node, and W_s' is the parameter matrix.

The feature interp(M_s') is projected back to the original space and is multiplied with Y. The channel dimension of $\widetilde{YX}$ is then converted to D dimension by 1×1 convolution. The purpose is to enhance the original features with fused features and extract semantic information. The equation is as follows:

$$\widetilde{YX} = \psi\left(Y_s \times \text{interp}\left(M_s'\right)\right). \tag{10}$$

The GCN_F in Graph Decoder follows the same form as in Graph Encoder. $\widetilde{X}$ and Y are dimensionally reduced and then multiplied to construct a new feature $M_{\mathcal{H}}'$, which combines feature maps from multiple channels from two different sources, $\widetilde{X}$ and Y. Then, inference is performed using graph convolution in the feature interaction space. The formula can be expressed as

$$M_{\mathcal{H}}' = \alpha(\widetilde{X})\beta(Y_s), \quad M_{\mathcal{H}}' \in \mathbb{R}^{D_2 \times D_1} \tag{11}$$

$$H_f' = \left(I + A_{\mathcal{H}}'\right) M_{\mathcal{H}}' W_{\mathcal{H}}'. \tag{12}$$

The decoded feature H_f' is projected back to the original space with Y as the reference, given by

$$\overline{YX} = \psi\left(Y_s \times H_f'\right), \quad \overline{YX} \in \mathbb{R}^{H \times W \times D}. \tag{13}$$

The final decoding result of Graph Decoder P is given by

$$P = Y + \widetilde{YX} + \overline{YX}. \tag{14}$$

D. Knowledge Review Module

The KRM is a multibranch structure. It enhances knowledge complementarity among layers by capturing information about the differences among network layers through conflicting and reversed attention [53]. The KRM is set between three key layers of the network, the input of which includes the backbone network output feature X, the graph convolutional encoder $\widetilde{X}$, and the graph convolutional decoder P. The detailed implementation is shown in Fig. 3.

Conflict attention (CoA) obtained diversity information from two different network layers L_1, L_2. The equation is as follows:

$$\text{CoA} = L_1(1 - L_2) + L_2(1 - L_1). \tag{15}$$

Reverse attention (ReA) uses low-level features to compensate for the lack of detail in high-level features. ReA converts the changed areas, which are predicted by the high-level features, into unchanged areas. Thus, forcing the network to pay more attention to the unchanged areas. The formula is as follows:

$$\text{ReA} = (1 - L_2). \tag{16}$$

The KRM inputs the fused features k_1, k_2, k_3 of the three layers into the CoA branch, the residual branch, and the ReA branch, respectively, for capturing the changed regions that are differentiated among the layers. Channel attention is also added to each branch to enhance the feature representation. The details are as follows:

$$k_1, k_2, k_3 = \text{Conv}_{1\times1}(\text{cat}(L_1, L_2)) \tag{17}$$

$$k_1^r = \text{Conv}_{3\times3}(\text{CA}(k_1 + k_1 \times CoA)) \tag{18}$$

$$k_2^r = \text{Conv}_{3\times3}(\text{CA}(k_2)) \tag{19}$$

$$k_3^r = \text{Conv}_{3\times3}(\text{CA}(k_3 + k_3 \times \text{ReA})). \tag{20}$$

The result of KRM, $K_{L_2 \to L_1}$, can be expressed as

$$K_{L_2 \to L_1} = \text{Conv}_{3\times3}(\text{cat}(L_1, k_1 + k_2 + k_3)) \tag{21}$$

where L_2 represents high-level features and L_1 represents low-level features relative to L_2 and $K_{L_2 \to L_1}$ denotes the formation of complementary knowledge from L_2 to L_1 through the KRM module.

The final result of the CF-GCN network is shown as

$$P^1 = \text{Conv}_{1\times1}\left(\text{cat}\left(K_{\widetilde{X} \to X}, K_{P \to \widetilde{X}}\right)\right) \tag{22}$$

where $K_{\widetilde{X} \to X}$ denotes the knowledge review from the graph convolutional encoder $\widetilde{X}$ to the backbone network output feature X. $K_{P \to \widetilde{X}}$ denotes the knowledge review from the graph convolutional decoder P to $\widetilde{X}$ knowledge review. Two KRMs are used to prevent misclassified information from propagating between different layers

$$\text{Pre} = \sigma\left(\text{upsample}\left(|P^1 - P^2|\right)\right) \tag{23}$$

where Pre is the change detection prediction result. P^1, P^2 represent the bitemporal image decoding features, $\sigma(\cdot)$ is the activation function of RELU, and upsample$(\cdot)$ is the upsampling operation with a rate of 4.

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IV. Experiments

A. Experiment Preparation

1) Datasets: To evaluate our approach, extensive experiments were conducted on two datasets, LEVIR-CD [18] and WHU-CD [52]. LEVIR-CD is a classic dataset for remote sensing image change detection, consisting of 637 pairs of high-resolution (0.5 m) images with 1024 × 1024 pixels in size. These bitemporal images are from 20 different areas in several cities in Texas, USA. They are collected between 2002 and 2018. The dataset was sliced into images in size of 256 × 256, resulting in training sets with 7120 pairs, validation sets with 1024 pairs, and test sets with 2048 pairs. WHU-CD is a public dataset for building change detection, consisting of a pair of high-resolution (0.075 m) remote sensing images in size of 32 507 × 15 354 pixels. It was divided into 7434 images in size 256 × 256. 5947 pairs are randomly selected for the training, 744 pairs for the validation, and the remaining 743 pairs for testing.

2) Evaluation Metrics: For the measurement of similarity between the predicted change pictures and ground truth labels, we introduced five quantitative evaluation metrics. They are $F1$-score ($F1$), precision (Pre.), recall (Rec.), intersection over union (IoU), and overall accuracy (OA) [14]. The definitions are as follows:

$$F1 = \frac{2}{\text{Rec.}^{-1} + \text{Pre.}^{-1}} \tag{24}$$

$$\text{Pre.} = \text{TP}/(\text{TP} + \text{FP}) \tag{25}$$

$$\text{Rec.} = \text{TP}/(\text{TP} + \text{FN}) \tag{26}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{27}$$

$$\text{OA} = \text{TP} + \text{TN}/(\text{TP} + \text{FP} + \text{FN} + \text{TN}) \tag{28}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative, respectively.

B. Experimental Environment

The network was implemented on PyTorch, and all experiments were conducted on NVIDIA A10. Common data augmentation techniques, including scaling, flipping, and cropping, were applied to the input images to increase the diversity of the training data. In particular, we used the image stylization unification [54] when preprocessing the WHU-CD dataset. Cross-entropy loss was chosen as the loss function, and stochastic gradient descent (SGD) was used as the optimizer. The initial learning rate and batch size were set to 0.01 and 16. The number of iterations was set to use pretraining epoch = 200 and training from scratch [55] epoch = 300 to ensure the fairness of the experiment. These hyperparameters were selected based on empirical evaluation and fine-tuning.

C. Analysis of Experimental Results

To validate the effectiveness of CF-GCN, several representative methods in change detection were compared with it, including pure convolution methods (FC-EF [36], FC-Siam-Di [36], FC-Siam-Conv [36], and ChangeStar [39]), Transformer + CNN methods (BIT [14], ICIFNet [38] and DMINet [40]), Transformer + MLP + multiscale-based method (ChangeFormer [26]), and graph convolution method (BGINet [31]). These methods were selected because they represent the SOTA in change detection and cover a range of different architectures and techniques. The performance of each method on LEVIR-CD and WHU-CD datasets was evaluated using the aforementioned evaluation metrics. The details are shown in Table I.

The specific analysis of the experimental results is as follows.

1) Results on WHU-CD Dataset: The results of comparative experiments are shown in Table II. It can be seen that our network has a remarkable advantage when compared with other networks in Pre. (94.81%), $F1$ (91.83%), IoU (84.90%), and OA (99.37%). In particular, comparing the $F1$ and IoU scores of the second-place DMINet [40], we have 0.6% and 1.03% improvement. It is a significant breakthrough in the pixel-level prediction task, demonstrating that our proposed CF-GCN achieves SOTA on WHU-CD.

There are many "non-semantic changes" in the WHU-CD dataset. Some visualization results are shown in Fig. 4. In Fig. 4(b), (d), and (f), most of the networks recognize the false-positive areas (red areas) as "non-semantic changes," and only CF-GCN do not misidentify them. In Fig. 4(c), all networks recognize concrete floors and ground objects as changed regions, but our network recognizes the lowest percentage of "non-semantic change" regions. This confirms that our network has a great advantage over the other networks in Table II in terms of detection metrics. In Fig. 4(a) and (e), the changed labels are regions with jagged edges, and most of the networks fail to recognize the changed region in (e). Only the CF-GCN is able to recognize it successfully and retained the changed region with jagged edges. Not only the changed regions, but also CF-GCN can identify the boundary regions of small samples without sticking or merging with other regions in Fig. 4(h). This demonstrates the ability of CF-GCN for detecting internal changes with complex structures.

2) Results on LEVIR-CD Dataset: The evaluation metric results are shown in Table III. CF-GCN is better on the results comparing to other networks, especially in $F1$ (90.96%), IoU (83.41%), and OA (99.09%). In particular, when comparing the $F1$ and IoU scores with the second-place ChangeStar [39], CF-GCN has an improvement of 0.24% and 0.39%. It illustrates that our proposed CF-GCN obtains satisfactory results in terms of comprehensive metrics.

The visualization results are shown in Fig. 5. In Fig. 5(a), most of the networks are successful in identifying the change regions and only some of them are inaccurate in identifying the boundary regions. In Fig. 5(c)–(e), there are false-positive and false-negative regions in all networks due to the high interference of "non-semantic changes" such as tree shadows. In particular, in Fig. 5(c), all other networks have large false negative errors, and our network recognizes the smallest percentage of incorrect regions. In Fig. 5(b) and (f), due to the difference in exposure levels when acquiring the bi-temporal images, the change in color is easily misinterpreted as a "changed" pixel. Compared with other networks, CF-GCN recognized all the changed houses and preserves the boundary

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TABLE I
DETAILED INTRODUCTION OF ALL COMPARATIVE EXPERIMENTS

FC-EF[36]	Image-level method that directly inputs concatenated bitemporal images into a fully connected network (FCN).
FC-Siam-Di[36]	Feature-level method that encodes bitemporal images into a Siamese network within FCN to obtain feature difference information, and then decodes the multi-scale features by concatenating them and inputting them into the FCN decoder.
FC-Siam-Conv[36]	Feature-level method that encodes bitemporal images into a Siamese network within FCN to obtain feature aggregation information, and then decodes the multi-scale features by concatenating them and inputting them into the FCN decoder.
BIT[14]	Feature-level method that maps the obtained features into token sequences for merging. The obtained token sequences are input into a Transformer for encoding, and then the resulting sequences are split and input into the Transformer decoder along with the ResNet network output features. For fair comparison experiments, BIT_ResNet18 and BIT_ResNet50 were also included for comparison.
ChangeFormer[26]	Feature-level method that extracts features from bitemporal images by inputting them into a Siamese Transformer, and then obtains multi-scale features that are differentially merged into an MLP module.
ICIFNet[38]	Feature-level method. The local features and global features, extracted by CNN and Transformer, respectively, are interactively communicated at the same spatial resolution. This communication is facilitated by a linearized Conv Attention module.
BGINet[31]	Feature-level method. By leveraging the concept of non-local operations and mapping the features obtained from the backbone network to the graph structure space, BGINet uses a unified self-focus mechanism for bitemporal images. It can enhances the information coupling between the two temporal images while effectively suppressing task-irrelevant interference.
ChangeStar[39]	Feature-level method. The ChangeMixin module is proposed to be applied to the fully supervised change detection task by taking the dual temporal feature maps as inputs and then connecting them in two different temporal alignments along the channel axis, providing an inductive bias in the network architecture.
DMINet[40]	Feature-level method, by unifying self-attention (SelfAtt) and cross-attention (CrossAtt) in a single module, uses an intertemporal joint-attention (JointAtt) block to steer the global feature distribution of each input, then motivats information coupling between intra level representations and meanwhile suppressing the task-irrelevant interferences.

TABLE II
COMPARISON RESULTS ON WHU-CD TEST SETS. ALL EVALUATION METRICS ARE EXPRESSED AS PERCENTAGES (%). COLOR CONVENTION: BEST, SECOND-BEST, AND THIRD-BEST

Model	Params.(M)	Pre.	Rec.	F1	IoU	OA
FC-EF[36]	1.35	86.12	70.76	77.6 9	63.51	98.39
FC-Siam-Di[36]	1.35	86.37	75.71	80.69	67.62	98.56
FC-Siam-Conc[36]	1.54	86.53	83.08	84.77	73.56	98.82
BIT_Resnet18[14]	3.45	77.70	89.70	83.27	71.33	98.57
BIT_Resnet50[14]	11.40	76.81	**91.01**	83.31	71.40	98.55
Changeformer[26]	41.02	**90.23**	83.49	86.73	76.57	98.98
ICIFNet[38]	23.82	86.20	90.27	88.19	78.87	99.04
BGINet[31]	2.88	87.88	91.27	**89.54**	**81.07**	**99.15**
ChangeStar[39]	33.96	82.41	93.33	87.53	77.83	98.94
DMINet_Resnet18[40]	6.24	93.00	89.52	91.23	83.87	99.32
CF-GCN	13.58	94.81	89.04	91.83	84.90	99.37

features of the houses. In Fig. 5(g), a large number of houses appear and disappear, and all networks misjudge one of the houses, recognizing the "non-semantic change" caused by the change of light in different seasons as a change region. Despite

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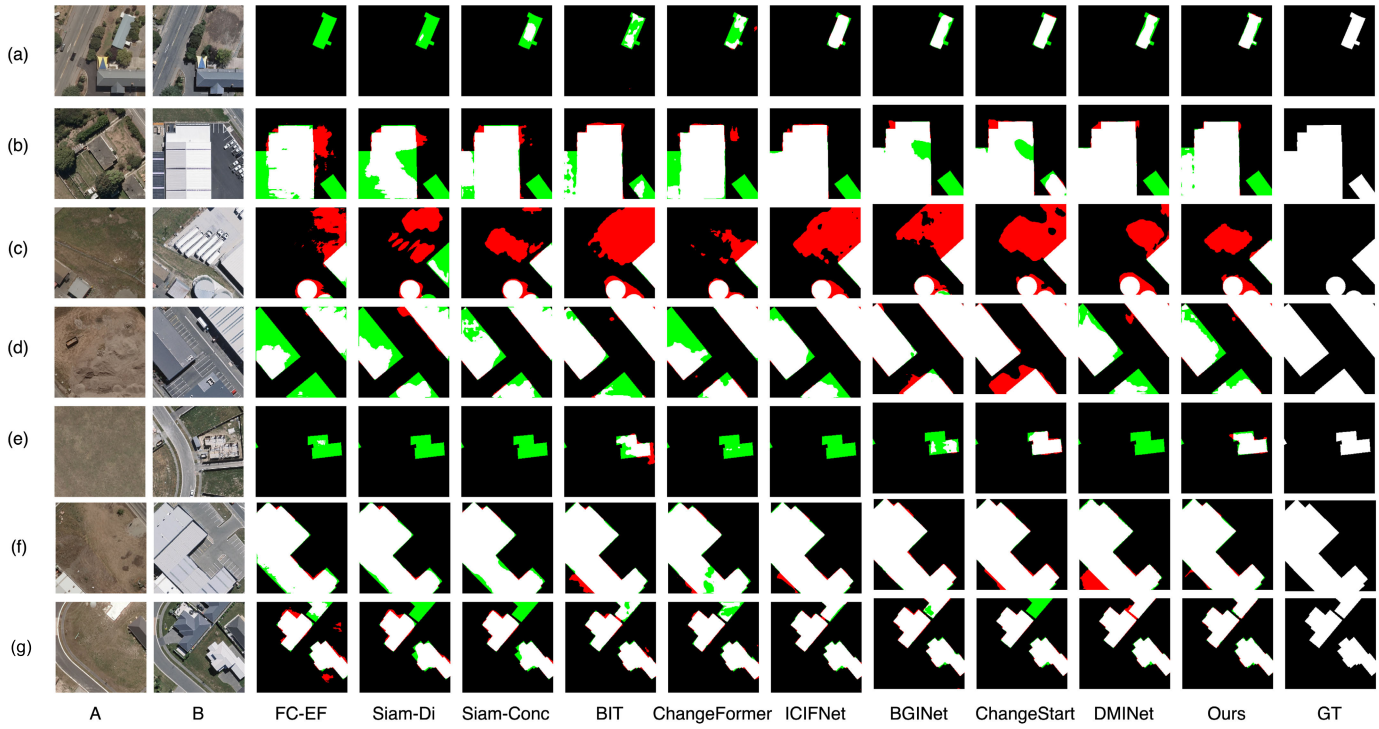


Fig. 4. Visual results of different models on WHU-CD. (a)–(g) Ten representative samples. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative. A, B are images of bi-temporal images. GT is the ground truth.

TABLE III

COMPARISON RESULTS ON LEVIR-CD TEST SETS. ALL EVALUATION METRICS ARE EXPRESSED AS PERCENTAGES (%). COLOR CONVENTION: BEST, SECOND-BEST, AND THIRD-BEST

Model	Params.(M)	Pre.	Rec.	F1	IoU	OA
FC-EF[36]	1.35	87.72	81.46	84.47	73.12	98.47
FC-Siam-Di[36]	1.35	90.69	83.25	86.82	76.71	98.71
FC-Siam-Conc[36]	1.54	90.30	84.59	87.35	77.54	98.75
BIT_Resnet18[14]	3.45	90.87	87.71	89.26	80.61	98.92
BIT_Resnet50[14]	11.40	92.00	84.94	88.33	79.10	98.85
Changeformer[26]	41.02	90.48	83.80	87.01	77.01	98.72
ICIFNet[38]	23.82	90.76	89.45	90.09	81.97	99.00
BGINet[31]	2.88	89.84	89.35	89.60	81.15	98.94
ChangeStar[39]	33.96	**90.69**	**90.81**	**90.75**	**83.06**	**99.06**
DMINet_Resnet18[40]	6.24	**91.79**	**89.67**	**90.72**	**83.02**	**99.07**
CF-GCN	13.58	**91.75**	**90.18**	**90.96**	**83.41**	**99.09**

the challenges, CF-GCN achieves visualization results that are very close to ground truth.

3) Performance Analysis: In Fig. 6, we analyze the trade-off between IoU scores and computational costs of various classical and recently proposed methods (e.g., Changer, BIT, ChangeFormer, and so on) on two datasets. The results show that although the above methods prove to perform well, they usually have significant computational overhead. On the other hand, baseline models like FC-EF require minimal computational resources, but the accuracy is low. Our method outperforms the baseline model in terms of accuracy while still maintaining an acceptable computational cost. In a word, the effectiveness and efficiency of our method in remote sensing image change detection tasks are ensured.

In Fig. 7, for non-semantic changes, such as tree shadows, artificial roads, cars, and color changes produced by light, our network can identify them completely. For the artificially constructed buildings in the fourth row, comparing with the DMINet, we only misjudged the objects on the ground, not a large area. CF-GCN has a clear advantage over other networks in analyzing and processing the semantic information of remote sensing.

D. Ablation Experiments

1) Selection of Backbone Network: In CNNs, shallow feature maps contain rich texture and color information, while deep feature maps contain rich semantic information. Ablation

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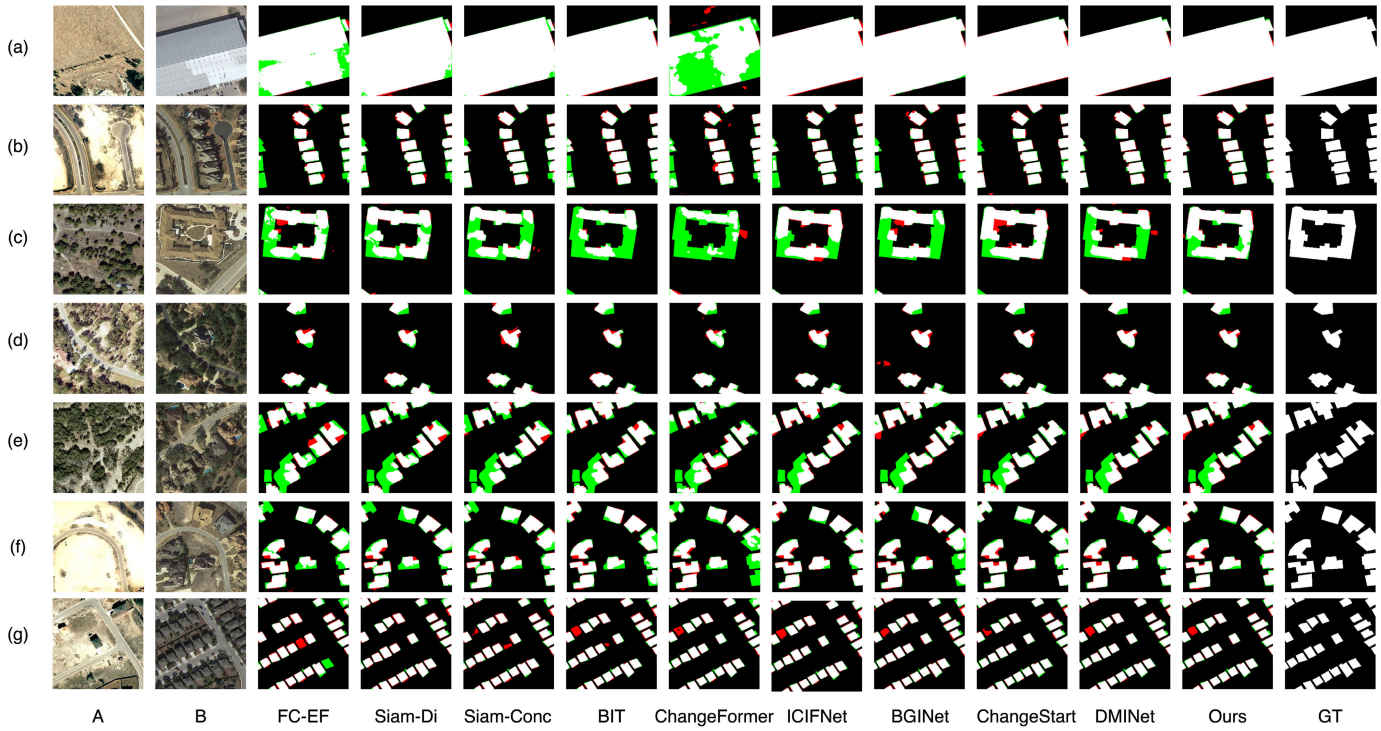


Fig. 5. Visual results of different models on LEVIR-CD. (a)–(g) Ten representative samples. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative. A, B are images of bi-temporal images. GT is the ground truth.

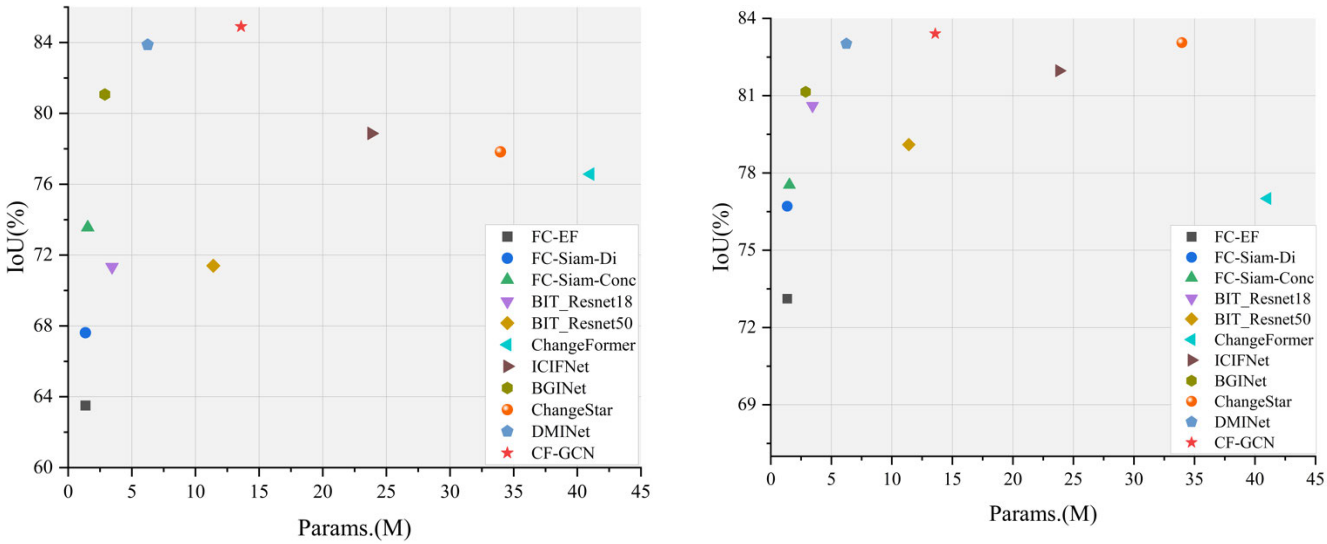


Fig. 6. Comparing the complexity of different methods in terms of Params.(M) and IoU-score on the (left) WHU-CD dataset and (right) LEVIR-CD dataset.

experiments are conducted to study the effect of network depth on the GCN Encoder and GCN Decoder, in order to verify the model's ability to aggregate semantic information from feature maps. The experimental results are shown in Table IV.

The results show that when choosing ResNet50 as the backbone, CF-GCN performs better than the other backbone on both datasets. Therefore, ResNet50 is chosen as the backbone of this article to obtain richer semantic information from deeper layers.

2) Input Image Size: To verify the processing capability of the GCN for different input size images, feature maps of three different sizes (16 × 16, 32 × 32, and 64 × 64) are used in CF-GCN. The Graph Encoder and Graph Decoder modules are used to encode and decode feature maps of different sizes, and the results are shown in Table V. On the LEVIR-CD dataset, as the feature map size increases, CF-GCN can better distinguish between real changes and irrelevant changes in larger feature maps. When compared with processing image size of 16 × 16, the computational cost of processing image size of 32 × 32 increases by 13% while an 8.5% (IoU) improvement is achieved. When processing an image size of 64 × 64, the computational cost increases by 21.56%, but there is no improvement in accuracy. Therefore, 32 × 32 is chosen as the input feature map size for CF-GCN.

3) Module Effectiveness: Experiments are conducted on both datasets to verify the effectiveness of the proposed core

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TABLE IV

ABLATION EXPERIMENTS ON THE IMPACT OF NETWORK DEPTH ON GRAPH CONVOLUTIONAL FEATURE EXTRACTION ABILITY. ALL EVALUATION METRICS ARE EXPRESSED AS PERCENTAGES (%)

		LEVIR-CD		WHU-CD	
Model	**Backbone**	**F1**	**IoU**	**F1**	**IoU**
CF-GCN	ResNet18	90.44	82.55	90.95	83.47
CF-GCN	ResNet34	90.67	82.93	91.47	84.28
CF-GCN	ResNet50	**90.96**	**83.41**	**91.83**	**84.90**
CF-GCN	ResNet101	90.53	82.70	90.75	83.06

TABLE V

ABLATION EXPERIMENTS ON THE ABILITY OF GRAPH CONVOLUTION TO HANDLE IMAGE SIZE. ALL EVALUATION METRICS ARE EXPRESSED AS PERCENTAGES (%)

			LEVIR-CD		WHU-CD	
Model	**Size**	**FLOPs(G)**	**F1**	**IoU**	**F1**	**IoU**
CF-GCN	16	38.67	85.65	74.91	87.49	77.76
CF-GCN	32	43.93	**90.96**	**83.41**	**91.83**	**84.90**
CF-GCN	64	47.01	90.74	83.05	90.59	82.81

TABLE VI

ABLATION EXPERIMENTS ON THE GCN_C, GCN_F, BPM, AND KRM, SHOWING $F1$ AND IOU SCORES ON THE TEST SETS OF LEVIR-CD AND WHU-CD. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

							LEVIR-CD		WHU-CD	
Model	**Encoder**	**Decoder**	**GCN_C**	**GCN_F**	**BPM**	**KRM**	**F1**	**IoU**	**F1**	**IoU**
Baseline							88.48	79.33	83.58	71.80
CF-GCN	√	√		√			88.33	79.09	89.69	81.33
CF-GCN	√	√	√				90.19	82.14	90.55	82.73
CF-GCN	√		√	√			90.15	82.07	90.97	83.43
CF-GCN	√	√	√	√			90.30	82.31	91.29	83.97
CF-GCN	√	√	√	√	√		90.47	82.59	91.47	84.27
CF-GCN	√	√	√	√		√	90.73	83.03	91.63	84.55
CF-GCN	√	√	√	√	√	√	**90.96**	**83.41**	**91.83**	**84.90**

modules in change detection tasks. The core modules include GCN_C, GCN_F, BPM, and KRM permutations into CF-GCN. The experimental results are shown in Table VI. When there are only GCN_F, the effect on the LEVIR-CD dataset is not significant compared to the baseline. The reason is that the GCN_F is designed to perform feature interactions between channels, the processing of small changes is mainly realized by the GCN_C. Using the GCN_F alone is not effective enough for recognizing small object changes in the LEVIR-CD dataset. When both GCN_C and GCN_F are added to the network, the performance on the LEVIR-CD dataset is improved by 2.48% ($F1$) and 4.08% (IoU) compare to the baseline. And on WHU-CD dataset, it is improved by 7.71% ($F1$) and 12.16% (IoU). It demonstrates that the graph convolution architecture can improve the performance of change detection networks. After adding both BPM and KRM, about 1.1% and 0.93% (IoU) boosts are again obtained on both datasets. It is demonstrated that texture characterization of shallow networks coupled with timely retrospective learning of knowledge helps the change detection network to obtain complete and regular prediction results.

TABLE VII

ABLATION EXPERIMENTS ON THE INTERNAL STRUCTURE OF THE GCNDECODER. ALL EVALUATION METRICS ARE EXPRESSED AS PERCENTAGES (%)

		LEVIR-CD		WHU-CD	
Model	**Construction**	**F1**	**IoU**	**F1**	**IoU**
CF-GCN	$X \rightarrow$Y	**90.96**	**83.41**	**91.83**	**84.90**
CF-GCN	$Y \rightarrow$X	90.44	82.54	91.49	84.32

4) GCN Decoder Module Structure: In order to verify the validity of the internal structure of the decoder module, we set up a set of ablation experiments. The experimental results are shown in Table VII. The identities of $\widetilde{X}$ and Y [as shown in Fig. 1(d)] in the decoder are exchanged separately to verify the impact of their projection position differences. The results show that it works better when projecting $\widetilde{X}$ back to the Y-based space than the opposite. Therefore, we chose to project the fusion back to the original feature in the GCN Decoder.

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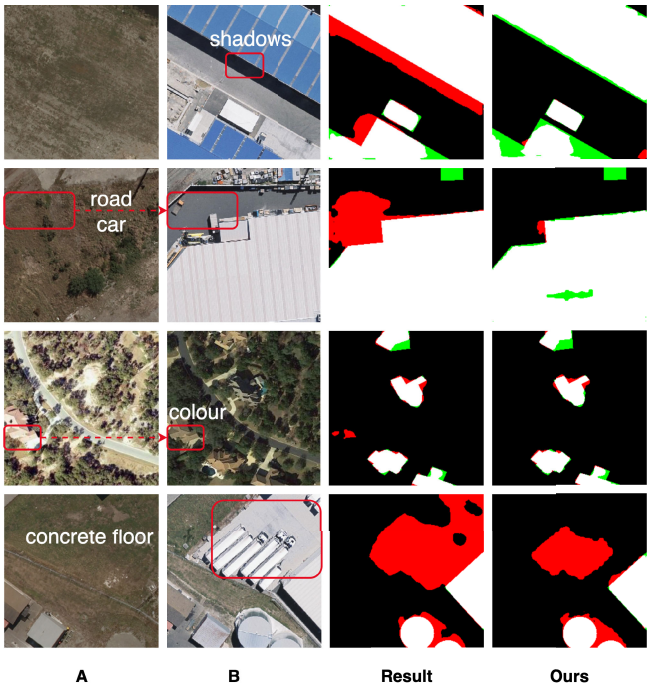


Fig. 7. Variety of "non-semantic changes" affects change detection results. A, B are images of bi-temporal images. The results, in order from top to bottom, are Siam-Di, Bit, ChangeFormer, and ChangeStar. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

V. Conclusion

This article proposes a remote sensing image change detection network CF-GCN based on graph convolution structure. By constructing GCN_C and GCN_F, semantic information in the scene is modeled over long distances, identifying the interfering factors "non-semantic changes" in bi-temporal images accurately. At the same time, the BPM and KRM are introduced. The BPM is added to allow Encoder and Decoder learning more boundary features in shallow networks during the training process, identifying changes in boundary areas of images thereby. The KRM focuses on solving the problem of error knowledge propagation at different layers in deep networks. CF-GCN has been tested on two classic remote sensing image change datasets, LEVIR-CD and WHU-CD, and achieved SOTA results comparing with some other networks. With the development of technology, more graph convolution-based methods would appear in the field of remote sensing image change detection. In our future work, we would like to improve this method for conducting unsupervised or weakly supervised change detection task.

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